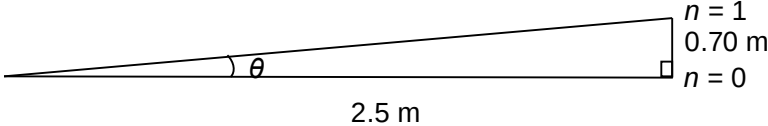


5(a)(i)	Wavelength = 12 m $f = \frac{v}{\lambda} = \frac{1.25}{12}$ = 0.104 Hz	[1] sub
(ii)	Distance between the two buoys = (1.25)(2 × 60) = 150 m Phase difference = (150 / 12) × 2π = 25π rad which is equivalent to π rad OR Path difference = (1.25)(2 × 60) = 150 m = 12.5 λ Phase difference = π rad	[1] distance [1] ans
(b)(i)	Diffraction is the spreading of waves through an aperture or around an obstacle.	[1]
(ii)	From Fig. 5.3, separation of first order from central maximum for grating B = 0.70 m  $\tan \theta = \frac{0.70}{2.5} \Rightarrow \theta = 15.64^\circ$ Using $\sin \theta = n\lambda p$ $\sin 15.64^\circ = (1)(590 \times 10^{-9})p$ $\therefore p = 4.57 \times 10^5 \text{ m}^{-1}$ Note: can also use the 2nd and 3rd order separations: separation of 2nd order from central maximum = 1.6 m separation of 3rd order from central maximum = 3.5 m	[1] working to find θ [1] sub [1] ans
MC	Ecf within part	

5(b)(iii)	The separation of the orders is observed to be smaller in the vertical spread. This shows that the angle of diffraction due to grating B is smaller. Since $d \propto \frac{1}{\sin \theta}$ (from $d \sin \theta = n\lambda$), slit spacing for diffraction grating B is larger than that of diffraction grating A. OR The highest order that can be observed for any diffraction grating depends on the relationship $n \leq \frac{d}{\lambda}$. Since there are more orders observed in the vertical spread, this implies that diffraction grating B has a larger slit separation than diffraction grating A,	[1] reason [1] deductn
(iv)	Any one of the following: - Approximately equal spacing of the diffraction fringes (or Equal spacing of fringes in the horizontal spread and the vertical spread) - More number of spots seen on the screen. (or Spacing between fringes decreases) - The intensity of higher order (2nd or 3rd order) fringes of the nylon mesh is approximately the same as that of the 1st order. - It would not be a sharp and distinct fringe pattern as the nylon does not have a sharp edge as that of a diffraction grating.	[1]
6(c)	It is the phenomenon whereby electrons from a metal are emitted when	[1]